D2: Dual-Source $q(\mathbf{F})$ Update Under $\eta = \mathbf{YF}\tilde{\beta}$

Cluster B (Cox-on-YF Reformulation) — Derivation D2 (Primary)

Cross-reference: `derivations/Modular_UpdateDerivations/` q(L) section (n↔p symmetry)

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1 Context and Goal

Under the Cluster B reformulation $\eta_i = (\mathbf{y}_i\mathbf{F})\,\tilde{\boldsymbol{\beta}}$, the factor matrix $\mathbf{F} \in \mathbb{R}^{p\times K}$ appears in *both*:

1. the genomics likelihood $\mathbf{Y} \approx \mathbf{L}\mathbf{F}^\top$, and
2. the Cox PH likelihood through $\eta_i = \sum_k (\sum_j y_{ij} f_{jk})\,\tilde{\beta}_k$.

This is analogous to how $\mathbf{L}$ had a dual role in the original model. The derivation follows the same structure as the original $q(\mathbf{L})$ derivation (`Modular_UpdateDerivations/` q(L) section), with the n- and p-dimensional roles swapped.

Goal of D2. Derive the factor-wise ELBO for $q(f_{jk})$, identify the dual-source EBNM inputs $A_{F,j}$ and $B_{F,j}$ for both sources, and establish that the scale imbalance between survival and genomics is resolved under this formulation.

2 Model Recap

Generative model.

$$y_{ij} \mid \mathbf{l}_i, \mathbf{f}_j, \tau_j \;\sim\; \mathcal{N}\!\left(\mathbf{l}_i^\top \mathbf{f}_j,\; \tau_j^{-1}\right), \qquad h_i(t) = h_0(t)\, e^{\eta_i}, \quad \eta_i = \sum_k \underbrace{\left(\sum_j y_{ij} f_{jk}\right)}_{(z_F)_{ik}} \tilde{\beta}_k.$$

Notation.

- $\mathbf{Z}_F = \mathbf{Y}\mathbf{E}[\mathbf{F}] \in \mathbb{R}^{n\times K}$: projection scores held fixed per outer CAVI iteration.
- $w_i = W_{ii}$: negative diagonal Hessian weights from the Cox Taylor expansion.
- $z_{\text{no-}k,i} = z_i - \sum_{k'\neq k} (z_F)_{ik'}\, \mathbb{E}[\tilde{\beta}_{k'}]$: partial working response with factor k removed.
- $R_k = \mathbf{Y} - \sum_{k'\neq k} \mathbb{E}[\mathbf{l}_{k'}]\mathbb{E}[\mathbf{f}_{k'}]^\top$: partial genomics residual.

CAVI update frequency. $\mathbf{Z}_F$ is recomputed once per outer CAVI iteration (before the Cox Taylor expansion). Within the k-loop, $(z_F)_{ik} = \sum_j y_{ij}\,\mathbb{E}[f_{jk}]$ — so after $\mathbb{E}[\mathbf{f}_k]$ is updated, $\mathbf{Z}_F$ is stale. However, the standard Gauss-Seidel contract holds $\mathbf{Z}_F$ fixed for the duration of the inner k-loop; it is refreshed at the start of the next outer iteration. This matches how η is treated in the original model.

3 Factor-wise ELBO for $q(f_{jk})$

In Gauss-Seidel CAVI, we update $q(\mathbf{f}_k)$ holding all other quantities fixed.

3.1 Genomics Contribution

Expanding the squared error for gene j, factor k:

$$\mathbb{E}_q[\log p(\mathbf{Y} \mid \mathbf{L}, \mathbf{F}, \boldsymbol{\tau})]\Big|_{j,k} = -\frac{\tau_j}{2} \sum_i \mathbb{E}_q[(r_{k,ij} - l_{ik}\, f_{jk})^2] + \text{const}$$

$$= -\frac{\tau_j}{2} \left[\mathbb{E}_q[f_{jk}^2] \sum_i \mathbb{E}_q[l_{ik}^2] - 2\, \mathbb{E}_q[f_{jk}] \sum_i \mathbb{E}_q[r_{k,ij}] \mathbb{E}_q[l_{ik}] \right] + \text{const.} \tag{1}$$

Reading off the coefficient of $\mathbb{E}_q[f_{jk}^2]$ and $\mathbb{E}_q[f_{jk}]$:

$$A_{F,\text{gen},j} = \tau_j \sum_i \mathbb{E}_q[l_{ik}^2], \tag{2}$$

$$B_{F,\text{gen},j} = \tau_j \sum_i \mathbb{E}_q[r_{k,ij}]\, \mathbb{E}_q[l_{ik}] = \tau_j\, [\mathbf{R}_k^\top \mathbb{E}[\mathbf{l}_k]]_j. \tag{3}$$

3.2 Survival Contribution

The Cox PH likelihood enters via the Taylor-linearised surrogate. The relevant expected log-likelihood (up to constants in $q(\mathbf{f}_k)$) is

$$\mathbb{E}_q[\log p(\mathbf{t}, \boldsymbol{\delta} \mid \mathbf{Y}, \mathbf{F}, \tilde{\boldsymbol{\beta}})] \approx \sum_i \Big[u_i\, \eta_i - \tfrac{1}{2} w_i\, \eta_i^2 \Big] + \text{const}, \tag{4}$$

where u_i is the score residual and the term in η_i^2 is the Taylor quadratic.

Substituting $\eta_i = \sum_{k'} (z_F)_{ik'}\, \mathbb{E}[\tilde{\beta}_{k'}]$ and isolating the contribution from f_{jk} (through $(z_F)_{ik} = \sum_{j'} y_{ij'} f_{j'k}$):

Isolating f_{jk} in the survival term

Write $\eta_i = y_{ij}\, f_{jk}\, \mathbb{E}[\tilde{\beta}_k] + \tilde{\eta}_{\text{no-}j,i}$ where $\tilde{\eta}_{\text{no-}j,i}$ collects all terms that do not depend on f_{jk}. Then (4) contributes, for gene j, factor k:

$$-\tfrac{1}{2} \sum_i w_i\, \mathbb{E}_q[\eta_i^2]\Big|_{f_{jk}}$$

$$= -\tfrac{1}{2} \mathbb{E}_q[f_{jk}^2]\, \mathbb{E}[\tilde{\beta}_k^2] \sum_i w_i\, y_{ij}^2 + \mathbb{E}_q[f_{jk}]\, \mathbb{E}[\tilde{\beta}_k] \sum_i w_i\, z_{\text{no-}k,i}\, y_{ij} + \text{const}, \tag{5}$$

where $z_{\text{no-}k,i}$ is the partial working response ($\eta_{\text{no-}k,i} + u_i/w_i - \sum_{j'} y_{ij'} \mathbb{E}[f_{j'k}] \mathbb{E}[\tilde{\beta}_k]$ simplifies to $z_i - (z_F)_{\cdot,-k}\, \mathbb{E}[\beta_{-k}]$, cancelling the contribution of factor k).

Reading off the coefficients:

$$A_{F,\text{surv},j} = \mathbb{E}[\tilde{\beta}_k^2] \sum_i w_i\, y_{ij}^2 = \mathbb{E}[\tilde{\beta}_k^2]\, [\text{diag}(\mathbf{Y}^\top \text{diag}(\mathbf{w}) \mathbf{Y})]_j, \tag{6}$$

$$B_{F,\text{surv},j} = \mathbb{E}[\tilde{\beta}_k] \sum_i w_i\, z_{\text{no-}k,i}\, y_{ij} = \mathbb{E}[\tilde{\beta}_k]\, [\mathbf{Y}^\top (\mathbf{w} \circ \mathbf{z}_{\text{no-}k})]_j. \tag{7}$$

Correction (2026-10-15)

Equation (7) is incomplete. Removing only the other programs ($z_{\text{no-}k}$) leaves the other genes of program k in the residual, contrary to the definition of $\tilde{\eta}_{\text{no-}j}$ above. The complete term subtracts $\mathbb{E}[\tilde{\beta}_k^2] \sum_i w_i \, y_{ij} \big((z_F)_{ik} - y_{ij}\mathbb{E}[f_{jk}]\big)$ and makes the update a sequential sweep over genes. See `derivations/cox_on_YF/canonical_ZF/`. Results with $\alpha_F = 0$ do not use this term and are unaffected.

3.3 Combined EBNM Inputs

Mixing the two sources with weight $\alpha \in [0, 1]$:

$$A_{F,j} = (1 - \alpha)\, A_{F,\text{gen},j} + \alpha\, A_{F,\text{surv},j}, \tag{8}$$

$$B_{F,j} = (1 - \alpha)\, B_{F,\text{gen},j} + \alpha\, B_{F,\text{surv},j}. \tag{9}$$

EBNM call for $q(f_k)$ under Cluster B

Inputs (p-vectors):

$$x_{F,j} = \frac{B_{F,j}}{A_{F,j}}, \qquad s_{F,j} = \frac{1}{\sqrt{A_{F,j}}}.$$

Note on τ_j cancellation: In the pure-genomics limit ($\alpha = 0$), $x_{F,j} = B_{F,\text{gen},j}/A_{F,\text{gen},j}$, and τ_j cancels in $x_{F,j}$ (divides numerator and denominator equally) but *not* in $s_{F,j}$ (which has $A_{F,\text{gen},j}^{-1/2} \propto \tau_j^{-1/2}$). This is the same τ-cancellation property as in the original $q(\mathbf{F})$ derivation. When $\alpha > 0$, τ_j does not cancel globally (survival and genomics A terms have different τ dependencies), but numerically the expression in (8) is well-defined.

Prior family: point-exponential (or as configured in `prior_LF`)

4 Scale Imbalance Audit

A key question: does the Cluster B formulation fix the scale imbalance that plagued the original $q(\mathbf{L})$ update?

4.1 Original Problem (for reference)

In the original model ($\eta = \mathbf{L}\boldsymbol{\beta}$), the $q(\mathbf{L})$ precision was:

$$A_{L,\text{gen}} = \sum_j \tau_j \, \mathbb{E}[f_{jk}^2] \approx p\, \bar{\tau}\, \mathbb{E}[f^2] \qquad \text{vs.} \qquad A_{L,\text{surv},i} = \lambda\, w_i\, \mathbb{E}[\beta_k^2] \approx O(1).$$

The ratio $A_{L,\text{surv}}/A_{L,\text{gen}} \approx 1/(p\,\bar{\tau}) \approx 1/2000$ for PDAC-scale data — survival cannot compete.

4.2 Cluster B Scaling for $q(\mathbf{F})$

Genomics precision.

$$A_{F,\text{gen},j} = \tau_j \sum_i \mathbb{E}[l_{ik}^2].$$

This sums over n patients ($n \approx 200$–500). Typical values: $\tau_j \approx 0.1$–10, $\sum_i \mathbb{E}[l_{ik}^2] \approx n\,\sigma_l^2 \approx n \cdot 0.01$. So $A_{F,\text{gen},j} \approx n\,\tau_j\,\sigma_l^2$.

Survival precision.

$$A_{F,\text{surv},j} = \mathbb{E}[\tilde{\beta}_k^2] \sum_i w_i \, y_{ij}^2.$$

Typical values: $w_i \approx 0.05$–0.2 (Cox weight), $y_{ij}^2 \approx 1$ (standardised expression), $\mathbb{E}[\tilde{\beta}_k^2] \approx 0.01$–$0.1$. So $A_{F,\text{surv},j} \approx n \, \bar{w} \, \bar{y}^2 \, \mathbb{E}[\tilde{\beta}^2] \approx n \cdot 0.1 \cdot 1 \cdot 0.05$.

Ratio.

$$\frac{A_{F,\text{surv},j}}{A_{F,\text{gen},j}} \approx \frac{\bar{w} \, \bar{y}^2 \, \mathbb{E}[\tilde{\beta}^2]}{\tau_j \, \sigma_l^2} \approx \frac{0.1 \cdot 1 \cdot 0.05}{1 \cdot 0.01} = 0.5.$$

Scale imbalance resolved

The n factors cancel in the ratio — both $A_{F,\text{gen}}$ and $A_{F,\text{surv}}$ scale with n. The ratio is $O(1)$ rather than $O(1/p)$. Survival and genomics contributions to $q(\mathbf{F})$ are now comparable in magnitude, which is the structural fix Cluster B was designed to achieve.

5 Identifiability Under $\eta = \mathbf{Z}_F \tilde{\boldsymbol{\beta}}$

Scaling ambiguity

Under $\eta_i = \sum_k (\sum_j y_{ij} f_{jk}) \tilde{\beta}_k$, there is a scaling ambiguity: scaling $f_{jk} \to c \, f_{jk}$ and $\tilde{\beta}_k \to \tilde{\beta}_k / c$ leaves η_i unchanged. This is equivalent to the $\mathbf{L}$–$\boldsymbol{\beta}$ scaling ambiguity in the original model.

The EBNM prior on $q(\mathbf{f}_k)$ breaks this ambiguity by imposing sparsity. Specifically:

- The point-exponential prior shrinks small $|f_{jk}|$ to zero and sets a characteristic scale via the estimated σ_k in the EBNM family. This anchors the scale of f_{jk}.
- Given the anchored scale of $\mathbf{F}$, $\tilde{\beta}_k$ is identified through the EBNM prior on $q(\tilde{\beta}_k)$ (point-normal with EBNM-estimated scale).

Practical consequence. No additional orthogonality or normalization constraint is needed. The Gauss-Seidel updates cycle between $q(\mathbf{F})$ and $q(\tilde{\boldsymbol{\beta}})$, and the joint EBNM shrinkage provides implicit identification. This is the same mechanism as the original model (where $q(\mathbf{L})$ and $q(\boldsymbol{\beta})$ are jointly identified through EBNM).

6 Train/Test Consistency

Train

Training:

$$\hat{\eta}_i = \sum_k (z_F)_{ik} \, \mathbb{E}[\tilde{\beta}_k] = (\mathbf{y}_i \, \mathbb{E}[\mathbf{F}]) \, \mathbb{E}[\tilde{\boldsymbol{\beta}}].$$

Test:

$$\hat{\eta}_i^{\text{test}} = (\mathbf{y}_i^{\text{test}} \, \mathbb{E}[\mathbf{F}]) \, \mathbb{E}[\tilde{\boldsymbol{\beta}}].$$

Identical formula. No matrix inverse, no OLS projection, no special-case logic. The learned $\mathbb{E}[\mathbf{F}]$ and $\mathbb{E}[\tilde{\boldsymbol{\beta}}]$ transfer directly.

7 Code Implications

`code/update_F.R` — `update_F_k()`

- **New arguments:** `w` (n-vector), `EBeta_k` (scalar), `EBeta2_k` (scalar), `YtWY_diag` (p-vector, pre-computed), `YtWz_no_k` (p-vector, pre-computed per k)
- **New computation:**

```
A_F_surv <- EBeta2_k * YtWY_diag       # p-vector: eq. (5)
B_F_surv <- EBeta_k  * YtWz_no_k       # p-vector: eq. (6)
A_F <- (1-alpha)*A_F_gen + alpha*pmax(A_F_surv, A_floor)
B_F <- (1-alpha)*B_F_gen + alpha*B_F_surv
```

- **Pre-computation in caller (`fit_modular.R`):**

```
YtWY_diag   <- colSums(sweep(Y^2, 1, w, "*"))         # once per outer iter
YtWz_no_k_k <- as.vector(t(Y) %*% (w * z_no_k))      # once per k
```

`code/update_F.R` — `update_F_all()` Add `w`, `EBeta`, `EBeta2`, `Y`, `z` to signature. Pre-compute `YtWY_diag` once per call; compute `YtWz_no_k` per k.

8 Verification Checklist

D2 Verification

1. $A_{F,\text{surv},j}$ formula matches (6): $\mathbb{E}[\tilde{\beta}_k^2] \cdot \sum_i w_i y_{ij}^2$. ✓
2. $B_{F,\text{surv},j}$ formula matches (7): $\mathbb{E}[\tilde{\beta}_k] \cdot \sum_i w_i z_{\text{no-}k,i} y_{ij}$. ✓
3. Scale ratio $A_{F,\text{surv}}/A_{F,\text{gen}} \approx O(1)$ (not $O(1/p)$). ✓
4. τ_j cancels in $x_{F,j}$ in the pure-genomics limit ($\alpha = 0$). ✓
5. When $\alpha = 0$: $A_F = A_{F,\text{gen}}$, $B_F = B_{F,\text{gen}}$ (reduces to original). (to test)
6. When `w`=0: $A_{F,\text{surv}} = 0$, $B_{F,\text{surv}} = 0$. (to test)
7. When `EBeta_k`=0: $B_{F,\text{surv}} = 0$. (to test)
8. All F tests pass including 6 new dual-source tests. (to implement)